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## Research Article

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# Advancing Wine Fermentation: Extended Kalman Filter for Early Fault Detection

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## Abstract

This work proposes an Extended Kalman Filter (EKF) state estimation approach for early detection of stuck and sluggish wine fermentations. The goal is to provide accurate information to enologists during fermentation to facilitate timely intervention and decision making. The study investigates the sensitivity of the fermentation process to various factors such as model parameters and initial conditions, especially for unmeasured nitrogen. It also shows how the estimation depends on meaningful sugar measurements, which are not available during the lag phase of fermentation. According to Monte Carlo simulations, the estimation algorithm was able to predict 95% of the problematic fermentations within the first few days. When initial nitrogen measurements are taken into account, a reliable prediction is available on the first day in 80% of the cases, justifying the additional cost. These results support the use of advanced control and monitoring methods in wine production and other alcoholic fermentation processes.

**Keywords:** Extended Kalman Filter, Fault Detection, Uncertainty Propagation, Wine Fermentation

## 1 Introduction

Alcoholic fermentation, which involves the conversion of grape sugars to ethanol and carbon dioxide by yeasts, is a crucial step in winemaking. This complex microbial process requires optimal conditions to complete successfully; otherwise, it can

lead to problematic fermentations with significant economic losses. Factors that affect fermentation include toxicity due to high ethanol, low pH, extreme temperatures, and the limitation of key nutrients such as nitrogen and oxygen ([Bisson, 1999](#)).

The most common problems encountered in industrial wineries are *stuck* fermentations, characterized by a high level of residual sugar remaining at the end, and *sluggish* fermentations, marked by a low rate of sugar consumption (Malherbe et al, 2007). These problems result in reduced productivity and, in the worst-case scenario, all of the raw materials are lost. These consequences highlight the need to develop techniques for early detection of problematic batches and investigate the best intervention strategies.

Online monitoring is seen as a promising method for improving the winemaking process, as it can support the enologist in decision making during fermentation and allows the adoption of novel control techniques (Sablayrolles, 2009). This requires, first of all, the implementation of sensors to measure variables related to fermentation rate such as density or CO<sub>2</sub> production.

There are several data-driven methods for monitoring in the process industry such as Principal Component Analysis (PCA), Neuro-Fuzzy Systems, and Artificial Neural Networks (ANN) (Kadlec et al, 2009). This class of methods requires a large amount of data for building black-box predictive models, but limited knowledge about the underlying process is needed. In contrast, model-driven methods are based on first principles equations describing the chemical and physical phenomena of the process (Jiang et al, 2021). Although these methods usually assume several simplifying conditions, they are useful for describing the internal variables of the process that are not directly measured. These two approaches can also be used together in the development of hybrid models that could result in better prediction and extrapolation using the same amount of training data (von Stosch et al, 2014). For instance, Luna et al (2023) developed a hybrid model that integrates a mechanistic model for alcoholic fermentation with a machine learning algorithm for predicting the extraction of polyphenols in red wines. Another example is Coleman et al (2007) which models the wine fermentation with Michaelis-Menten style kinetics, but uses polynomials for the reaction rate terms to fit the data.

Data-driven approaches for monitoring and fault detection of wine fermentation can be found in the literature. Urtubia et al (2007) employed

PCA and cluster analysis (K-means) of data from 24 industrial Cabernet Sauvignon fermentations to detect anomalous behaviors in advance, using data on sugar, alcohol and organic acids. In this case study, it was possible to predict the likelihood of the process ending normally using only data from the first three days of fermentation. In later studies, the authors compared the prediction provided by multiway PCA, multiway partial least squares, and ANN on the same data, achieving 100 % accuracy (Urtubia et al, 2012). Despite achieving good results, the methods rely on measurements that are difficult to acquire in an industrial setting, such as alcohol and organic acids, limiting their applicability.

Among model-driven methods, nonlinear observers and filters, such as the Extended Kalman Filter (EKF), are among the most popular (Lyubenova et al, 2021). In Bázquez-Martell et al (2021), the authors combine an EKF with an asymptotic observer for the estimation of biomass and ethanol from high-density yeast cultures using measurements of dissolved oxygen, CO<sub>2</sub> and glucose. The proposed observer was able to accurately predict the switch between different metabolic regimes, which is an important sign of overflow metabolism. Sipos (2020) develop an EKF for wine fermentation considering measurements of temperature and CO<sub>2</sub> to estimate biomass and sugar concentrations. However, different models were used for the decay and exponential phases, increasing the complexity of the implementation.

Kalman filtering is also used for *fault detection*, where the fault is incorporated into the model equation and estimated along with the states (Kordestani et al, 2021). For example, the process can be monitored for operational problems or component degradation. When coupled with some kind of uncertainty propagation, the EKF can also be used for *fault prognosis*, where the future status of the process is predicted (Akker-mans et al, 2018). When the EKF equations are used for propagation of the mean and covariance of the estimation, it corresponds to a linear approximation of the “true” uncertainty, which can be inaccurate for nonlinear systems (Landgraf et al, 2023). Nevertheless, it has a low computational load and its implementation is simple compared to other methods such as Monte Carlo simulations and polynomial chaos expansion.

This work proposes a monitoring strategy for wine fermentation with two components: an EKF for online estimation; and a predictor of faulty fermentations at the end of the batch. Unlike previous studies, the proposed methodology combines the advantages of EKF with a first-principles model to obtain a robust algorithm for monitoring and fault prediction in wine fermentation processes. The goal is to provide accurate information in real time for the winemaker to support his decision making. The proposed method, alongside the simulation studies, result in the following contributions:

- Application of EKF for early detection of problematic wine fermentations based on final sugar prediction;
- Accurate prediction available online using only temperature and density measurements.
- Analysis of the main measurements affecting the prediction of the final sugar concentration;
- Analysis of the performance and reliability of the proposed method based on Monte Carlo scenarios.

This document is organized as follows: in Section 2, the fermentation process and EKF for fault detection are explained in details; in Section 3.1 a case study on white wine fermentation is presented, followed by Monte Carlo scenarios on white and red wine in Section 3.2; in Section 3.2.1 the effects of measuring initial nitrogen is analysed; finally, the results are discussed in sections 3.3 and 4.

## 2 Methods

After some notation, the wine fermentation process and the dynamic model considered in this work are discussed in detail. Then, the proposed state estimation and fault detection algorithms are presented.

### 2.1 Preliminaries and Notation

A continuous dynamic nonlinear model can be represented by:

$$\begin{aligned}\dot{x} &= f_c(x(t), u(t), p) \\ y(t) &= h(x(t))\end{aligned}$$

which has dimensions  $x \in \mathbb{R}^{n_x}$  for the states,  $u \in \mathbb{R}^{n_u}$  for the control inputs,  $p \in \mathbb{R}^{n_p}$  for the parameters, and  $y \in \mathbb{R}^{n_y}$  for the measured outputs.  $f_c(\cdot)$  and  $h(\cdot)$  represent nonlinear functions for the states and outputs, respectively. This model can be transformed to discrete time by an integration method or by an ODE solver, resulting in:

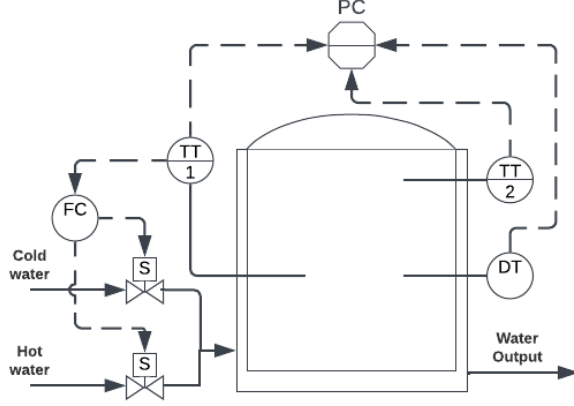
$$\begin{aligned}x(k+1) &= f(x(k), u(k), p) \\ y(k) &= h(x(k))\end{aligned}\tag{1}$$

where  $t = t_s k$  with  $t_s$  as the sampling time and  $k$  as the discrete sample. This notation for discrete-time variables can be abbreviated to  $x_k = x(k)$ . Finally, in the estimation context,  $\hat{x}$  represents the estimate of  $x$  and  $\hat{x}_{k|k-1}$  represents  $\hat{x}_k$  given the information at time  $k-1$ . For matrices,  $A'$  represents the transpose of  $A$ .

### 2.2 Wine Fermentation

The fermentation starts when the tank is inoculated with yeast, usually *Saccharomyces cerevisiae*, and ends when there is no sugar left, i. e. the wine is dry. This takes around 6-10 days operating around 25 °C for red wines and 12-20 days at 15 °C for white wines (Boulton et al, 1999). A diagram of the fermentation tank can be seen in Figure 1 which shows the temperature control using service water and sensors for temperature and density. Specifically for red wines, the juice is fermented in contact with the solids in a process called fermentative maceration which finishes when the must is pressed and the wine is separated (draining point). This process can continue long after the wine is dry, i. e. extended maceration, depending on the style of wine (Casassa and Harbertson, 2014).

The management of nitrogen additions at the beginning and during the fermentation process is key since it is the major limiting nutrient for yeast growth (Santamaria et al, 2020). A suitable dosage during the process helps to synthesize secondary metabolites such as esters that contribute to the aromatic wine quality (Seguinot et al, 2018). Furthermore, the amount of nitrogen in the must is an important factor on fermentation kinetics: musts with higher nitrogen content lead to faster fermentations, whereas lower concentration results in slower fermentations. Inadequate levels of nitrogen can cause stuck or sluggish fermentations.



**Fig. 1** P&ID of the tank fermentation. TT1 and TT2 are temperature sensors for different sections of the tank. DT is a density sensor

There are several sources of nitrogen available in grapes and musts, however, only some of them are easily assimilable by yeast such as ammonium and amino acids. Yeast assimilable nitrogen (YAN) is important because it is responsible for cell growth and metabolism, although it is expensive to measure accurately during fermentation (Bell and Henschke, 2005). In addition, there may be significant variability in YAN in vineyards, increasing the uncertainty of the process (Malherbe et al, 2007).

To ensure the success of wine fermentation, most industrial wineries have invested in automation technologies such as temperature control and automatic pump-overs, among others. In the last decade, affordable and reliable density sensors have been developed, which allow online monitoring of the fermentation progress (Sablayrolles, 2019). Density measurements based on differential pressure or production of  $\text{CO}_2$  can be used to infer the sugar content in the fermenting must (El Haloui et al, 1988). In this work, it is assumed that sugar ( $S$ ) and density ( $\rho$ ) are related by the following linear equation:

$$\rho = \frac{100}{250}S + 1000 \quad (2)$$

with sugar in  $g/L$  and density in  $kg/m^3$ . Furthermore, temperature ( $T$ ) is assumed constant and known.

## 2.3 Fermentation Model

Most wine fermentation models are described by a set of nonlinear ODEs with specific assumptions and purposes, as reviewed in Miller and Block (2020). In Coleman et al (2007), a simple lumped parameter model, assuming nitrogen as the primary limiting substrate and temperature-dependent kinetic parameters, was developed. This model is the one used in this work, with a few adjustments.

The continuous model equations are described by:

$$\dot{X} = \mu X - k_d E X, \quad \mu = \frac{\mu_{max} N}{K_N + N}, \quad (3a)$$

$$\dot{N} = -\mu \frac{X}{Y_{X/N}}, \quad (3b)$$

$$\dot{E} = \beta X, \quad \beta = \frac{\beta_{max} S}{K_S + S}, \quad (3c)$$

$$\dot{S} = -\beta \frac{X}{Y_{E/S}}, \quad (3d)$$

where the state variables are given by the biomass ( $X$ ), nitrogen ( $N$ ), ethanol ( $E$ ) and sugar ( $S$ ), while the other terms are given by the maximum specific growth rate ( $\mu_{max}$ ), maximum specific rate of sugar utilization ( $\beta_{max}$ ), half-velocity constant for nitrogen ( $K_N$ ), half-velocity constant for sugar ( $K_S$ ), death rate by ethanol ( $k_d$ ), the yield coefficient for cell mass grown per mass of nitrogen ( $Y_{X/N}$ ) and yield coefficient for ethanol produced per sugar consumed ( $Y_{E/S}$ ). These terms related to the kinetics, summarized in Table 1, are modeled as exponentials or polynomials up to order 2 dependant on temperature  $T$  or initial nitrogen  $N_0$ .

**Table 1** Model parameters for each term in (3)

| Parameter          | Polynomial              |
|--------------------|-------------------------|
| $\log \mu_{max}$   | $p_0 + p_1 T$           |
| $\log K_N$         | $p_2$                   |
| $\log k_d$         | $p_3 + p_4 T + p_5 T^2$ |
| $\log Y_{X/N}$     | $p_6 + p_7 N_0$         |
| $\log Y_{E/S}$     | $p_8$                   |
| $\log \beta_{max}$ | $p_9 + p_{10} T$        |
| $\log K_S$         | $p_{11}$                |

Source: Coleman et al (2007)

The nominal values for the polynomial coefficients and their standard deviations are shown in Table 2. The standard deviations for the parameters were taken from the 90 % credible region (CR) reported in Coleman et al (2007) divided by four, i. e. it is assumed that the distribution is normal and centered around the mean with  $\sigma = \text{CR}/4$ .

**Table 2** Nominal values ( $\bar{p}$ ) and standard deviations ( $\bar{\sigma}$ ) of the parameters

|          | $\bar{p}$  | $\bar{\sigma}$ |
|----------|------------|----------------|
| $p_0$    | -3.92      | 0.11           |
| $p_1$    | 7.82 e-2   | 0.45 e-2       |
| $p_2$    | -4.73      | 0.14           |
| $p_3$    | -9.81      | 0.42           |
| $p_4$    | -1.08 e-1* | 0.40 e-1       |
| $p_5$    | 4.78 e-3   | 0.86 e-3       |
| $p_6$    | 3.50       | 0.098          |
| $p_7$    | -3.61 e-3* | 0.36 e-3       |
| $p_8$    | -5.98 e-1  | 0.19 e-1       |
| $p_9$    | -2.98*     | 0.12           |
| $p_{10}$ | 7.71 e-2   | 0.55 e-2       |
| $p_{11}$ | 2.33       | 0.16           |

Source: Coleman et al (2007)

\*These values are different from the source since they appear to be typos.

## 2.4 Extended Kalman Filter

Extended Kalman filter (EKF), is a recursive algorithm that uses the past measurements and inputs to estimate the current state based on a statistical model. For estimation, the model given by (1) is augmented with noises on the state and output equations

$$\begin{aligned} x_{k+1} &= f(x_k, u_k, p, w_k) \\ y_k &= h(x_k) + v_k \end{aligned} \quad (4)$$

with  $w \in \mathbb{R}^{n_w}$  and  $v \in \mathbb{R}^{n_y}$  being random vectors modeled by zero-mean Gaussian noises with covariance matrices  $Q$  and  $R$ , respectively. For simplicity of the design, they are often modeled as non-correlated variables resulting in diagonal matrices containing the variance for each term. Furthermore,  $h(x_k)$  corresponds to the sugar measurement, which is simply the fourth state plus  $v_k$ .

Specifically for the fermentation model in (3), the noise  $w_k$  is modeled as an additive term in the parameters as in  $\bar{p} + w_k$ , where  $\bar{p}$  and  $Q$  defined by the values in Table 2. With this design choice, the model used for estimation in this work is as similar as possible as the one reported in Coleman et al (2007). It is common practice to include additive noise in the states when designing Kalman filters. However, they were not included in this work since it would require the addition of arbitrary tuning parameters for the EKF in the form of variance for each noise term.

The filter is initialized by two components: the expected mean  $\hat{x}_0$  and the expected covariance  $P_0$ . After that, the filter performs two steps at each sampling time: the *prediction* step and the *update* step. At first, an *a priori* state estimate is calculated by (4). At the same time, the state noise propagates through the model resulting in an *a priori* covariance  $P_{k|k-1}$ . The prediction equations are given by

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1}, \hat{p}) \quad (5a)$$

$$P_{k|k-1} = A_{k-1} P_{k-1|k-1} A'_{k-1} + G_{k-1} Q G'_{k-1} \quad (5b)$$

where  $A_{k-1}$  and  $G_{k-1}$  are linearization of the nonlinear model as in

$$A_{k-1} = \left. \frac{df}{dx} \right|_{\substack{\hat{x}_{k-1}, \\ u_{k-1}, \bar{p}}}, \quad G_{k-1} = \left. \frac{df}{dp} \right|_{\substack{\hat{x}_{k-1}, \\ u_{k-1}, \bar{p}}}.$$

Notice that  $G_{k-1}$  is a derivative to  $\bar{p}$  and not  $w_k$  since, in this work, the noise appears as  $\bar{p} + w_k$ . Furthermore, there is no time index in  $\bar{p}$  since it is assumed to be constant and equal to the nominal values.

Next, the update step is performed. First, the Kalman gain  $K_k$  is calculated using  $P_{k|k-1}$ ,  $C_k$  and  $R$ , where  $C_k$  is the linearization of  $h(\hat{x}_{k|k-1})$ . Then, using  $K_k$ , the estimates  $\hat{x}_{k|k-1}$  and  $P_{k|k-1}$  are updated according to the following equations:

$$K_k = P_{k|k-1} C'_k (C_k P_{k|k-1} C'_k + R)^{-1} \quad (6a)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (y_k - h(\hat{x}_{k|k-1})) \quad (6b)$$

$$P_{k|k} = (I - K_k C_k) P_{k|k-1} \quad (6c)$$

Real-world process constraints add additional challenges to the design of state estimators for nonlinear systems. For instance, all the states in

model (3) must be nonnegative. In the EKF algorithm, there are two steps which can result in invalid values: the prediction of  $\hat{x}_k$  in (5a), and the correction of the estimation in (6b). In this paper, it is used a simple strategy of clipping negative values to zero (Haseltine and Rawlings, 2005).

**Remark 1.** As can be seen in Table 1, the yield coefficient  $Y_{X/N}$  depends on the initial value of  $N$ . Hence, for a proper use of the model, the EKF should include online estimation of initial nitrogen, i. e.  $\hat{N}_{0|k}$ . Nevertheless, in order to simplify the design of the filter, this value is fixed at  $\hat{N}_{0|0}$  in this work.

## 2.5 Uncertainty Propagation

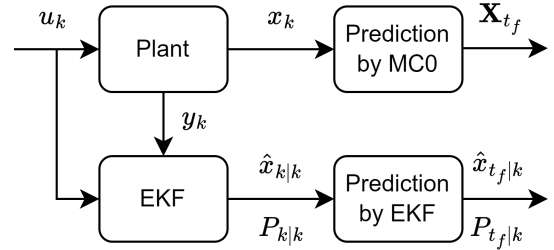
One of the challenges in nonlinear systems is that there is no easy way to propagate the covariance matrix  $P$  through the model. The EKF attempts to solve this using the nonlinear model for the estimate of  $\hat{x}$  and using a linearized model to calculate the uncertainty matrix  $P$  and the Kalman gain  $K$  at each sampling time, as can be seen in (5b) and (6c). Because of the linearized model for the covariance, EKF might have poor performance for highly nonlinear systems (Haseltine and Rawlings, 2005). Nevertheless, the EKF is a straightforward algorithm to generate predictions of the states  $\hat{x}_{t_f|k}$  and their uncertainties  $\hat{P}_{t_f|k}$  by sequentially applying (5) until  $t_f$ .

Although useful, EKF is not accurate for estimation or uncertainty propagation, specially for severe nonlinearities in the model. Therefore, a more accurate method, such as Monte Carlo (MC) simulations, is required for validation (Akkermans et al, 2018). For the fermentation model, the uncertainty at each sampling time is given by the covariance matrix  $P_{k|k}$  for the current state estimate  $\hat{x}_{k|k}$ , and the covariance matrix  $Q$  for the parameter vector  $p$ . Using the MC method, a large enough number of scenarios ( $M$ ) are generated with  $\hat{x}_{k|k}$  and  $p$  as central points, and  $P_{k|k}$  and  $Q$  as covariance matrices. Next, each scenario given by a set of  $x_k^i$  and  $p^i$ , is simulated by ODE solver until  $t_f$  resulting in  $\mathbf{X}_{t_f} = [x_{t_f}^0, x_{t_f}^1, \dots, x_{t_f}^M]$ , while statistical properties can be calculated such as the median, mean  $\mu_{t_f} = E[\mathbf{X}_{t_f}]$  or covariance  $P_{t_f} = \text{cov}(\mathbf{X}_{t_f}, \mathbf{X}_{t_f})$ . Then, this information can be compared to the result of the EKF algorithm in order to assess its accuracy.

A variation of the MC method applied in this work considers only the pair  $(p, Q)$  for the scenario generation and assumes knowledge of the current state  $x_k$ . This method, referred here as MC0, captures the uncertainty in the prediction caused only by the parameters of the model, removing the effects due to the estimation error of  $\hat{x}_k$ .

## 2.6 Proposed Method for Fault Prediction

The proposed method for monitoring and fault prediction, visualized in Figure 2, considers an EKF as an online observer and uses its prediction equations for uncertainty propagation. At each sampling time, the EKF uses the current measurement  $y_k$  and the input  $u_k$  to update the prediction for  $\hat{x}_{k|k}$  and its uncertainty  $P_{k|k}$ . With this information, the prediction equations for EKF are repeated until  $t_f$  resulting in  $\hat{x}_{t_f|k}$  and  $P_{t_f|k}$ . Consequently, at each sampling interval  $t_s$ , it is possible to evaluate the probability of a failure occurring at  $t_f$  from those values. At the same time, for validation, the MC0 method is applied to the  $x_k$  considering only the uncertainty in  $p$ . The result from MC0 represents the best achievable performance for the EKF since it removes the effects of estimation error in  $\hat{x}_k$ .



**Fig. 2** Block diagram summarizing the proposed method for monitoring and fault prediction based on the EKF, and MC0 running in parallel for validation

Using the information provided by the above methods, it is possible to evaluate, at each  $t_s$ , the likelihood of a fault occurring in the future. For the purpose of this paper, which focuses on stuck and sluggish fermentations, a fault is defined as a sugar concentration at  $t_f$  above a desired value. For example, a fault could be defined as  $S(t_f) \geq 2 \text{ g/L}$  which corresponds to a dry wine, or could be higher for musts that are drained before

the end of alcoholic fermentation, which is common for varietal wines (Luna et al, 2021). With this definition, an alarm for a possible fault could be raised at any time when the prediction is above the threshold.

### 3 Results and Discussion

Simulation studies written in Python are performed to evaluate the performance of the proposed method for wine fermentation. The software CVODES is used as the ODE solver interfaced by CasADi for simulation and for calculating the sensitivities required by the EKF (Hindmarsh et al, 2005; Andersson et al, 2019).

The first case study focuses on the performance of the EKF for a specific scenario by considering the nominal parameter values. The second case study is composed of numerous MC scenarios where the overall performance of the algorithm is evaluated. Then, the MC scenarios are repeated considering measurement for initial nitrogen. Finally, the results are discussed in further detail in the last section.

#### 3.1 Illustrative Scenario

This simulation study, implements the EKF for fault detection in the nominal case for a white wine scenario with validation of the prediction by the MC0 method.

The sampling time used is  $t_s = 6$  h with  $t_f = 240$  h or 10 days. The desired sugar concentration at  $t_f$  is 10 g/L with a final value higher than that representing a problematic fermentation. The fermentation temperature of 15 °C is constant and known. It is assumed that measurement for sugar is available, and that it is disturbed by a zero-mean Gaussian noise with  $\sigma_y = 5^2$ . The initial values for the states are 0.2 g/L for biomass,  $N_0 = 150$  mg/L,  $S_0 = 265$  g/L and zero for ethanol. Both the system and the estimator have the same nominal parameters given by  $\bar{p}_i$  and  $\bar{\sigma}_i$  (Table 2). This scenario results in  $S(t_f) = 26.4$  g/L which is a sluggish fermentation.

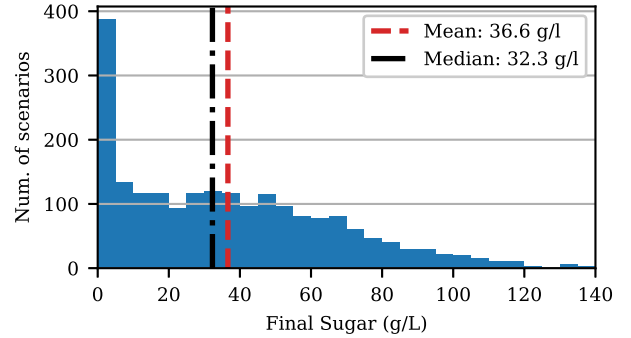
Meanwhile, the initial estimate for the EKF is the same as the process with the exception of nitrogen, which is significantly different with

$\hat{N}_0 = 150$  mg/L. The initial uncertainty  $P_0$  considered in the estimator is given by

$$P_0 = \text{diag}(0.025^2, 0.050^2, 0, 5^2), \quad (7)$$

with units in grams per liter. Notice that the standard deviation of 50 mg/L for nitrogen in the second term is large considering the nominal value, and  $\sigma$  for ethanol is zero, since it is certain that there is no alcohol at  $t_0$ .

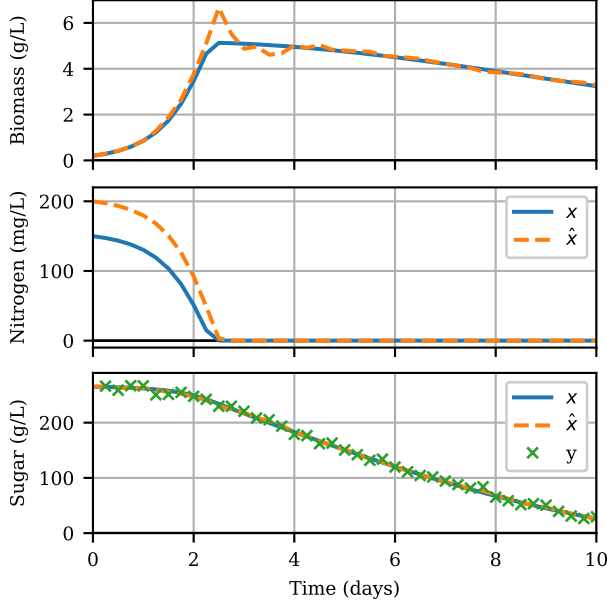
Figure 3 shows an histogram of the prediction of the sugar at the end of the fermentation process,  $S(t_f)$ , using the MC0 method with  $M = 2049$ . Although MC0 does not consider the uncertainty in the states, the simulated scenarios result in a wide range for  $S(t_f)$ , showing that this system is highly sensitive to the parameters. Most of the scenarios end successfully with almost no sugar left, but there is a significant tail which increases the mean and median. The histogram is not bell-shaped and, because of that, the standard deviation and the mean of the scenarios may be biased.



**Fig. 3** Histogram of  $S(t_f)$  using the MC0 method for the nominal model with  $n_{MC} = 2049$  and no uncertainty in  $x(0)$ . For this scenario,  $S(t_f) = 26.4$  g/L

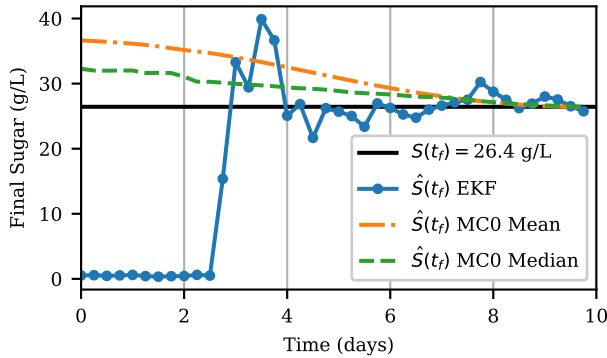
Figure 4 shows the estimation of the states by the EKF. At first, there is a large error in initial nitrogen, but after the lag phase of the fermentation is over, the filter is able to correct the estimated states, resulting in an error close to zero in steady-state.

Figure 5 shows the prediction of final sugar  $S(t_f)$  by the EKF and MC0 method during the 10 days of fermentation. The EKF starts with an optimistic initial estimate of 200 mg/L for nitrogen



**Fig. 4** Estimation of the states by EKF for a white wine scenario resulting  $S(t_f) = 26.4$  g/L. Estimation for ethanol is omitted for brevity

and, because of that, the prediction for  $\hat{S}(t_f)$  is zero for the first 60 hours. After the lag phase of the fermentation ends and the measurement for sugar starts changing, the filter quickly updates the prediction. This sudden increase in the predicted value for  $S(t_f)$  way above the threshold of 10 g/L could be interpreted as an alarm of problematic fermentation. After a few more iterations, the prediction converges to the final value around 26 g/L confirming the problematic fermentation.



**Fig. 5** Prediction of the sugar at  $t_f$  by EKF and by MC0 (median and mean) compared to the final value of 26.4 g/L

Figure 5 also shows two values for the predictions by MC0: the mean and the median. It can be seen that the mean of the MC scenarios is less reliable since the distribution is not normal, resulting in an overprediction of the final sugar. Therefore, in the following sections, the mean is discarded for MC0 in favor of the median.

Overall, the MC0 method using the median is more accurate compared to the EKF. However, it still has a considerable error in  $S(t_f)$  in the first few days, even though no uncertainty is considered for  $x_k$ . This shows that the best achievable performance for the EKF, represented by MC0, can still be inaccurate. Despite that, the EKF was still able to generate an alarm in a reasonable time.

### 3.2 Monte Carlo Scenarios

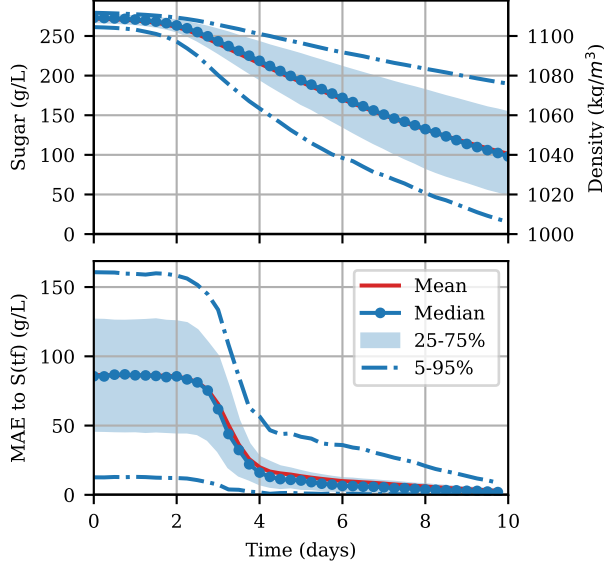
This section shows Monte Carlo analysis of the proposed EKF for two cases: White Wine (WW) and Red Wine (RW). Each scenario is simulated as described in Section 3.1, but using different initial conditions and parameters for the model. Each scenario is defined by the variables described in Table 3 with  $t_f = 10$  days for WW and 5 days for RW. The desired sugar at  $t_f$  for both cases is 10 g/L. For each scenario, the temperature is constant for the whole batch with 12-18 °C for whites and 22-28 °C for reds. Scenarios were generated using the Sobol method from SciPy with  $M = 1024$  plus the nominal scenario, totaling 2050 simulations for both cases. Meanwhile, the EKF is initialized with the same information: nominal  $N_0$  and  $\bar{p}$  from Table 3, and  $P_0$  from (7). For 2050 scenarios, 1026 scenarios presented faults considering the threshold of 10 g/L, with 876 being sluggish fermentation and 150 being stuck.

**Table 3** Variables ranges for each scenario

| Variable | Nominal   | Distribution         |
|----------|-----------|----------------------|
| $N_0$    | 200 mg/L  | Uniform [50, 350]    |
| $S_0$    | 265 g/L   | Uniform [250, 280]   |
| Temp. WW | 15 °C     | Uniform [12, 18]     |
| Temp. RW | 25 °C     | Uniform [22, 28]     |
| $p$      | $\bar{p}$ | Normal (see Table 2) |

Figure 6 shows the average sugar in evolution for faulty scenarios and the Mean Absolute Error (MAE) for the prediction of  $S(t_f)$  by the EKF algorithm for the WW case. On the bottom graph,

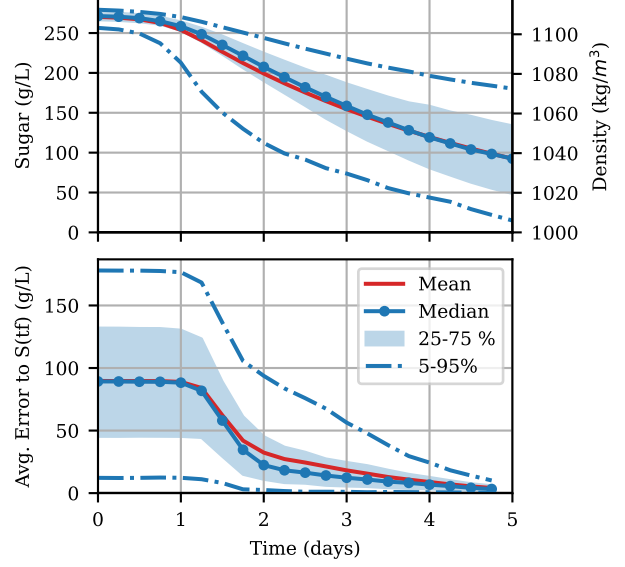
it can be seen that the prediction error for  $S(t_f)$  starts fairly high, but, around the third day, the error drops considerably. This happens around the same time that fermentation leaves the lag phase and starts the exponential phase with a faster conversion of sugar to ethanol. Since EKF depends heavily on sugar measurements, it is expected that it will only update its model after the process provides meaningful information, which happens after the fermentation leaves the lag phase.



**Fig. 6** White Wine case: Average sugar in time for scenarios with faults (top graph) and Mean Absolute Error (MAE) for the prediction of  $S(t_f)$  by the EKF (bottom graph)

The same can be seen for the RW case in Figure 7 with a decrease in average error around 36 to 48 hours. However, for this case, there is a larger range for the error compared to the WW case. This might be the result of a larger range for sugar (top graph) in faulty scenarios, or a shorter fermentation length.

For both cases, the prediction error is too large on the first days of fermentation in order to generate alarms, at least in this simulation study. Because of that, it is important to ignore the information before some minimum time has passed such as  $t_{min} = 24$  h for RW and 48 h for WW. With this criteria, the *first* alarms produced by EKF are summarized in Table 4.



**Fig. 7** Red Wine case: Average sugar in time for scenarios with faults (top graph) and Mean Absolute Error (MAE) for the prediction of  $S(t_f)$  by the EKF (bottom graph)

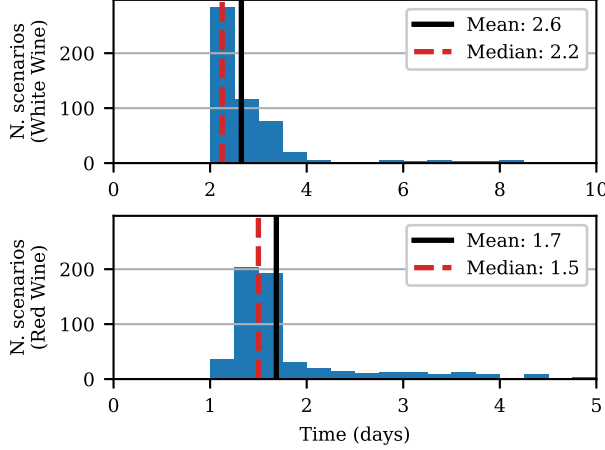
**Table 4** Confusion Matrix for the first alarm generated by EKF for both WW and RW cases

|           |             |             |
|-----------|-------------|-------------|
| Total:    | Alarm:      | No Alarm:   |
| 2050      | 1103        | 947         |
| Fault:    | True Pos.:  | False Neg.: |
| 1026      | 1010        | 16          |
| No Fault: | False Pos.: | True Neg.:  |
| 1024      | 93          | 931         |

The algorithm was successful in 1941 scenarios (94.7%), where 1010 were scenarios with faults and 931 with no faults or alarms. There were 93 false alarms, which is where the algorithm expects a fault that will not occur. However, since only the first one is analyzed in each scenario, it is expected that the following alarms were more accurate in the prediction. More concerning are the 16 scenarios where the EKF failed to generate an alarm for a faulty batch. In spite of that, the final sugar for these cases was close to the threshold with a median of 11.7 g/L and a maximum of 17.6 g/L.

Figure 8 shows histograms of the time of the *first* alarm generated by the EKF for each scenario. For the WW case, most alarms were generated in the first 2-3 days with few outliers. Meanwhile, for the RW case, most alarms were

generated between the first and second days with a heavier tail. This histogram, alongside the graph of error in time in Figure 7, shows that it is harder to predict the fermentation for RW probably because of the short duration. However, in both cases, this histogram shows that the algorithm is quite reliable in providing early alarms to the enologist in time for acting on the process.



**Fig. 8** Histogram of time of the first alarm generated by EKF in each scenario, separated by WW (top graph) and RW (bottom graph) cases

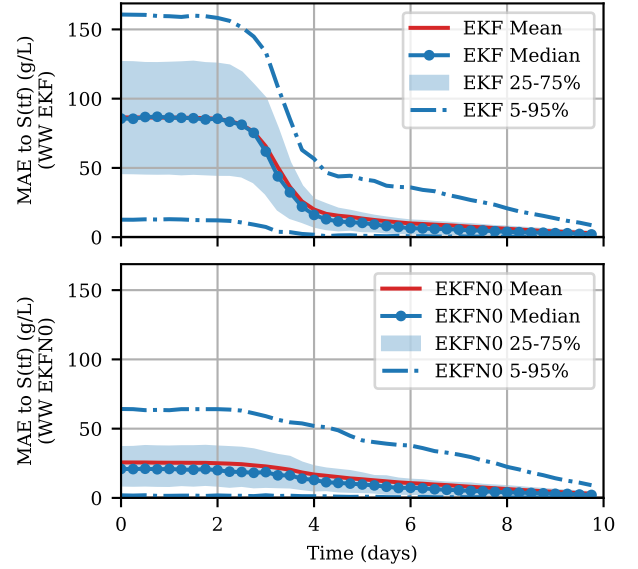
For example, in industrial wineries it is common to add nitrogen a second time to ensure a successful process in terms of the productivity and quality. This decision is carried out typically when the density reaches the range of  $1080\text{-}1050\text{ kg/m}^3$  that coincides with the time range of the first alarm generated (Luna et al, 2021). As can be seen in figures 6 and 7, the prediction error decreases significantly *before* reaching  $1080\text{ kg/m}^3$  which means that the information is reliable enough to be used when deciding how much nutrient should be added.

### 3.2.1 Measuring Initial Nitrogen

This section explores the effects of measuring nitrogen at the beginning of fermentation. All of the MC scenarios of the last section are repeated with two differences. First, the initial variance for nitrogen in  $P_0$  is reduced from  $0.050^2$  to  $0.015^2$  which corresponds to a higher confidence on  $\hat{N}_0$ . This represents a measurement error for nitrogen

around  $\pm 30\text{ mg/L}$  which is still significant considering the range for the variable. Second, at the beginning of each scenario,  $\hat{N}_0$  is initialized as  $N_0$  plus a zero-mean Gaussian noise. This new algorithm is named EKFN0.

The comparison between EKF and EKFN0 can be visualized in Figure 9 with the mean prediction error in time for WW. The addition measurement for  $N_0$  in EKFN0 results in a significant reduction in the prediction error for the first few days, improving the confidence of any alarms generated in this period. Furthermore, it can be seen that the reduction in the prediction error is not significant after the end of the lag phase (around 3-4 days). The results for the RW case are omitted for brevity since they are similar to the WW.



**Fig. 9** Mean Absolute Error (MAE) for the prediction of  $S(t_f)$  in time for WW in scenario with faults. EKF (top graph) vs. EKFN0 (bottom graph). The RW case shows similar results

The performance of EKFN0 in generating alarms is very similar to that of EKF, as reported in the Confusion Matrix of Table 4. Almost all of the scenarios with false positives and false negatives for EKF resulted the same for EKFN0. The main difference was the time of the first alarm for EKFN0. While the EKF gives the results shown in the histogram in Figure 8, EKFN0 generated 87% of the alarms for WW and 80% for RW

on the first day of fermentation, with a few outliers. As summarized in Table 5, the mean time of the first alarm generated was reduced in 73 % for WW and 53 % for RW. This is a significant improvement which could justify the investment in nitrogen measurement in some cases.

**Table 5** Time for the first alarm generated by EKF and EKFN0 for both WW and RW cases

| Case  |    | First Alarm (days) |        |
|-------|----|--------------------|--------|
|       |    | Mean               | Median |
| EKF   | WW | 2.6                | 2.2    |
| EKFN0 | WW | 0.7                | 0.0    |
| EKF   | RW | 1.7                | 1.5    |
| EKFN0 | RW | 0.6                | 0.0    |

### 3.3 Discussion

The case study on white wine in Section 3.1 showed that the EKF is able to adequately estimate unmeasurable states, such as active biomass. This can help the winemakers on deciding how to act on the process. For example, if the amount of biomass is not sufficient, it can be helpful to add more nitrogen so that it can grow. But if there is enough biomass but the fermentation is slow, maybe the temperature can be increased.

There are two main sources of uncertainty that lead to estimation error. First of all, there is the problem of uncertainty in the models' parameters that is inherent to the system. The histogram in Figure 3, with the predicted sugar concentration ( $S(t_f)$ ) at the end of fermentation, highlights the sensitivity of the system to the parameters. This aspect can only be improved by developing models that better represent the underlying phenomena and by performing experiments to collect data. This can reduce the uncertainty in the model and would improve its prediction accuracy.

The second aspect is the use of effective algorithms for estimation and prediction. It is known that EKF can show poor performance in some cases, especially for high nonlinearities and constraints (Haseltine and Rawlings, 2005). One reason for that is the use of a first-order method for uncertainty propagation which decreases accuracy and can lead to a poor estimation (Gustafsson

and Hendeby, 2012). Although the EKF performed well in this work, it is expected that more advanced methods for uncertainty propagation such as the sigma points or polynomial chaos expansion will lead to better estimation and fault prediction (Akkermans et al, 2018).

It is possible to use different methods for estimation and fault prediction. The EKF for estimation could be substituted by the Unscented Kalman Filter for better performance or by the Moving Horizon Estimation for the use of constraints, although the design process for these methods is more involved. Meanwhile, the uncertainty propagation for faults could be done by the MC method, with the downside of increased computational cost. These possibilities are worth exploring.

Even though the proposed algorithm can be improved, it performed well on the MC scenarios showing robustness under diverse conditions. The average sugar concentration profiles for faulty scenarios (Figures 6 and 7) highlighted the challenges of accurate prediction in the initial stages, particularly during the lag phase. The decrease in prediction error around 36 to 48 hours corresponded to the transition from the lag phase to the exponential phase, indicating the EKF's reliance on meaningful sugar measurements for effective updates. Still, it is early enough for the process to be acted upon.

The histogram of the time of the first alarm of each scenario (Figure 8) provided insight into the reliability of the EKF in providing early alarms. Most of the alarms are concentrated around the average which means that, if an alarm will be generated, it is expected to occur around the end of the lag phase. However, for shorter fermentations, more outliers are expected.

The first alarm generated by the algorithm is fairly accurate with 1010 scenarios (91.6 %) being true positive and 93 (8.4 %) false positive, as seen in Table 4. To further increase the reliability of these alarms, one option is to act on the system only after a few alarms are generated in sequence, which decreases the likelihood of a false alarm. Furthermore, the sampling time could be reduced so that more measurements are available. This happens because the EKF has more information available for a better estimation and prediction. Another important aspect is to use the uncertainty

$P_{t_f|k}$  alongside  $\hat{x}_{t_f|k}$  in the fault definition. With this, an alarm would only be generated when  $\hat{S}(t_f)$  is above a threshold and the uncertainty is low enough.

The last case study shows the benefits of measuring initial nitrogen. At least for the model considered, total nitrogen is crucial since it is directly related to peak biomass which, in turn, is necessary for successfully converting sugar to ethanol. Measuring nitrogen at the beginning, even with a fairly large error, improve the reliability of the proposed algorithm, although it leads to cost in materials and labor. However, the cost could be avoided by first running the proposed algorithm without measuring  $N_0$  and analyzing the results. If the prediction leads to a successful fermentation with a high degree of certainty, then the measurement could be skipped for that batch.

## 4 Conclusion

In this study, a monitoring strategy based on EKF for state estimation and early fault detection is proposed in order to provide accurate information during fermentation. This model-based method can support the enologist in detecting problematic batches and designing intervention strategies, while also enabling advanced automatic control which often requires knowledge of all the states of the process. The algorithm was tested in diverse simulation scenarios in order to assess performance and usefulness.

Analysis based on MC simulations showed how the process is sensitive to uncertainty on model parameters and initial conditions, especially for nitrogen. It also showed how the estimation is dependent on meaningful measurements for sugar which are not available during the lag phase of the fermentation. Despite these challenges, the main goal of generating accurate early alarms is achieved.

There are at least two aspects of the proposed algorithm which can be improved. First, more accurate models will result in better estimation, which leads to better fault prediction. Another one is the use of a more accurate uncertainty propagation method for faults. Further analysis could also indicate if effort should be concentrated on designing better estimation and prediction methods, or if it is better to gather more data for developing accurate models.

Apart from the technical aspect, it is important to improve the performance analysis of the algorithm since the study was conducted only on simulations. For example, the EKF could be applied to data from industrial batches to check if the conclusions drawn here still hold.

The presented findings contribute to the ongoing efforts in leveraging advanced control and monitoring techniques for optimizing wine fermentation processes (Sablayrolles, 2009). In an industrial setting, the estimation could be used alongside an automatic system for nutrient addition which would autonomously decide how much nitrogen to add in order to avoid problematic batches. It also enables the use of online model-based controllers such as predictive control. The proposed EKF approach, along with the conducted simulation studies, pave the way for future research in winemaking and analogous processes.

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